Gain PoC

Audio plugin proof of concept

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In one line: we built a small volume-control plugin that loads and works in a professional music app, proving we can deliver the client's real plugin.

What the client asked for

The client has an idea for an audio plugin and wants a developer who can build it in the three standard formats (**VST3**, **AU** and **AAX**). Before that, they asked for a *small proof of concept*: build a simple plugin, load it in **REAPER**, and check it in **Logic Pro**.

The plugin itself is deliberately simple. The real test is whether we can take a plugin from code all the way to something that works inside real music software.

What we built

Gain PoC is a single-knob volume plugin (from silent up to +12 dB).

- **Three versions from one codebase:** VST3 (REAPER, Cubase, Ableton…), AU (Logic, GarageBand) and a standalone app.
- **Runs on every modern Mac:** both Apple Silicon and Intel.
- **Behaves like a professional plugin:** smooth volume changes with no clicks, the knob can be automated, and settings are saved with the project.
- **Built with JUCE 8**, the framework most commercial plugins are built on.

How we know it works

Check	Result
Apple's official plugin test (`auval`), Apple Silicon and Intel	**Passed**
Industry stress test (`pluginval`), strictest level, both formats	**Passed**
Loaded in REAPER 7.82 as VST3 and AU	**Passed**
Knob set to −6 dB in REAPER → audio measured exactly 6 dB quieter	**Passed**

All test reports and audio files are in the GitHub repository.

Run it locally

You need: a Mac with Xcode, Homebrew and REAPER (60-day free trial).

1. Install the build tool:
   `brew install cmake`
2. Download the project:
   `git clone --recurse-submodules https://github.com/omverma1810/JUCE.git`
3. Build it (the first build takes a few minutes):
   `cd JUCE`
   `cmake -B build -G Xcode`
   `cmake --build build --config Release`

4. That's it. The plugin is installed automatically, and REAPER finds it on its next launch. The standalone app is in `build/GainPlugin_artefacts/Release/Standalone/`.

How to demonstrate it (about 5 minutes)

Screen-record this, with the system audio audible.

1. **Show the project:** the GitHub page and the test results above.
2. **Open REAPER:** drag any song or audio file onto a new track.
3. **Add the plugin:** click the track's **FX** button and search for *Gain PoC*. Pick the VST3 version.
4. **Play and turn the knob:** the volume changes smoothly as you turn it.
5. **Automate it:** show the track's envelopes, draw a volume curve for *Gain* and play it back. The knob moves by itself.
6. **Save and reopen:** save the project, close it and reopen it. The knob is where you left it.
7. **Repeat with the AU version**, to show both formats work.
8. **Optional:** open the standalone *Gain PoC* app and play your microphone through it.

About Logic Pro

Logic Pro is a paid app, but Apple offers a **free 90-day trial**. Two free alternatives:

- **GarageBand** (free on the Mac App Store) uses the same audio engine as Logic. If the AU works in GarageBand, it will work in Logic.
- **JUCE AudioPluginHost**, a free test app we have already built, loads both versions.

Next steps for the real plugin

- **Scope the client's idea:** effect or instrument, controls, and how the interface should look.
- **AAX (Pro Tools):** needs the client's Avid developer account and signing tools.
- **Distribution:** sign and notarize the plugin with Apple so it installs cleanly on any Mac, and package it as an installer.
- **Windows:** the same code can be built for Windows when needed.

Going further: Vibe FX

To make the demo more impressive, we built a second plugin, **Vibe FX**: an effects rack with the sounds people use in TikTok and YouTube song edits, plus a live dashboard.



- **11 effects:** speed (slowed / sped-up), lo-fi, filter, bass boost, drive, chorus, echo, reverb, 8D audio, stereo width, and intro/outro fades.
- **15 one-click presets:** Slowed + Reverb, Nightcore, Lo-Fi Chill, Vaporwave, 8D Audio, Telephone, Old Radio, Underwater, Echo Canyon, Bass Boosted and more.
- **Live dashboard:** see which effects are on, compare the original and processed sound on a spectrum display, and watch the waveform and levels.
- **Built-in player:** drop any song into the standalone app; no music software needed.
- **Tested:** passes Apple's official plugin test (auval), and all 15 presets were rendered in REAPER.

Demo in 3 minutes:

1. Double-click `demo/Start Vibe FX Demo.command`: the app opens with a demo song loaded. (Or open *Vibe FX* and drop any song onto it.)
2. Press **Play**, then step through the presets with the < > buttons.
3. Click **Bypass** to switch between the original and the effect.
4. Turn a few knobs and watch the spectrum and signal chain react.
5. In REAPER, open `demo/vibefx_presets/VibeFX_demo.rpp` to show it inside a DAW.

Pre-rendered examples of every preset are in `demo/vibefx_presets/`.